CNENGR202 Instrumentation and Control

Part 1 T1

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Example 1

- Compute the transfer function for the system of the first-order linear differential equation $\dot{y} + ky = u$, and find the output y for all time $(0 \leq t < \infty)$ when the input $u = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$

Example 1

Solution

$$u=\begin{cases} t & t\geq 0 \\ 0 & t<0 \end{cases} \longrightarrow U(s)=\frac{1}{s^2}$$

$$\dot{y}+ky=u \longrightarrow sY(s)+kY(s)=\frac{1}{s^2}$$

$$\downarrow$$

$$Y(s)=\frac{1}{s^2(s+k)}$$

Example 1

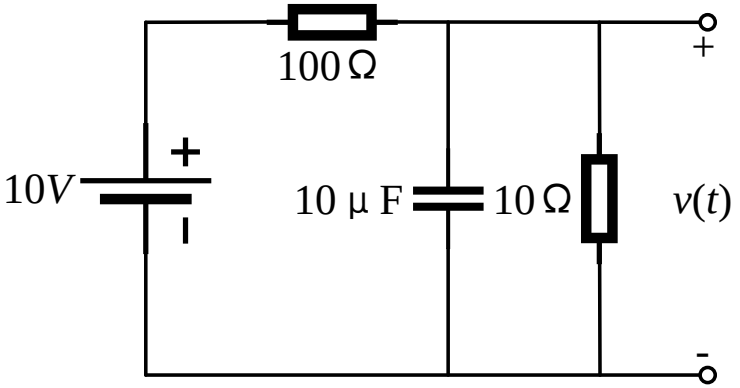
Solution $$Y(s)=\frac{1}{s^2(s+k)}$$

$$Y(s)=-\frac{1}{k}\frac{1}{s^2}+\frac{1}{k^2}\frac{1}{s}+\frac{1}{k^2}\frac{1}{s+k}$$

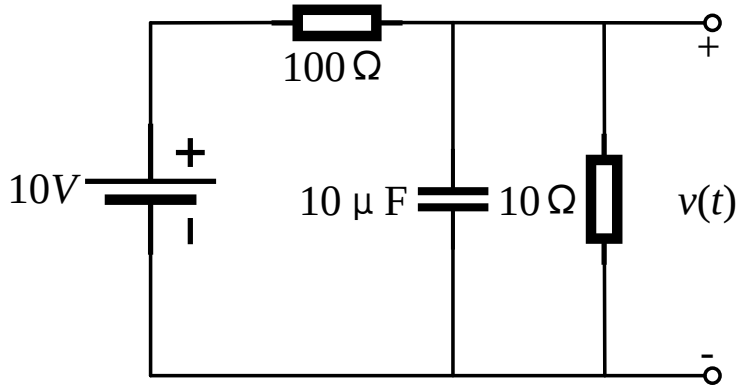
$$y(t)=-\frac{1}{k^2}+\frac{1}{k}t+\frac{1}{k^2}e^{-kt}$$

Example 2

- Compute the transfer function and $v(t)$ for the RC circuit driven by a 12V voltage source ($t > 0$) as shown in the right Fig.



Example 2



Solution

- Output function in frequency domain

$$v(s)=\frac{\frac{100000}{s+10000}}{100+\frac{100000}{s+10000}}\times 10=\frac{10000}{s+11000}$$

- Output function in time domain

$$v(t)=10000e^{-11000t}$$

Example 3

- A system is $\dot{y} + y = u$, determine the response to an input $u = \sin(2t)$ for $t > 0$

Example 3

Solution

$$u=\sin(2t) \longrightarrow U(s)=\frac{2}{s^2+4}$$

$$\dot{y}+y=u \longrightarrow Y(s)=\frac{1}{s+1}U(s)=\frac{2}{(s+1)(s^2+4)}$$

$$Y(s)=\frac{2}{(s+1)(s^2+4)} \longrightarrow Y(s)=\frac{0.4}{s+1}-\frac{0.4s}{s^2+4}-\frac{0.4}{s^2+4}$$

$$\downarrow$$

$$y(t)=0.4e^{-t}-0.4\cos(2t)-0.2\sin(2t)$$

Example 4

Find the final value of the signal corresponding to $Y(s) = \frac{2}{s(s-1)}$

Example 4

Solution

$$y(\infty) = \lim_{s \to 0} s \cdot \frac{2}{s(s-1)} = -2$$

$$Y(s) = \frac{2}{s(s-1)} \longrightarrow Y(s) = -\frac{2}{s} + \frac{2}{s-1} \longrightarrow y(t) = -2 + 2e^{t}$$

$$\lim_{t \to \infty} y(t) = \infty$$

The true final value is unbounded, because of the unstable pole at $s = 1$.

Example 5

Find the solution to $\ddot{y}(t)+5\dot{y}(t)+4y(t)=u(t)$ $y(0)=0, \dot{y}(0)=0$

$u=2e^{-2t}$ $t\geq 0$

Example 5

Solution

- Taking the Laplace transform of both sides, we get

$$s^2Y(s)+5sY(s)+4Y(s)=\frac{2}{s+2}$$

- Solving for $Y(s)$ yields

$$Y(s)=\frac{2}{(s+2)(s^2+5s+4)}=\frac{2}{(s+2)(s+1)(s+4)}$$

- The partial-fraction expansion using the cover-up method is

$$Y(s)=-\frac{1}{s+2}+\frac{2/3}{s+1}+\frac{1/3}{s+4}$$

Example 5

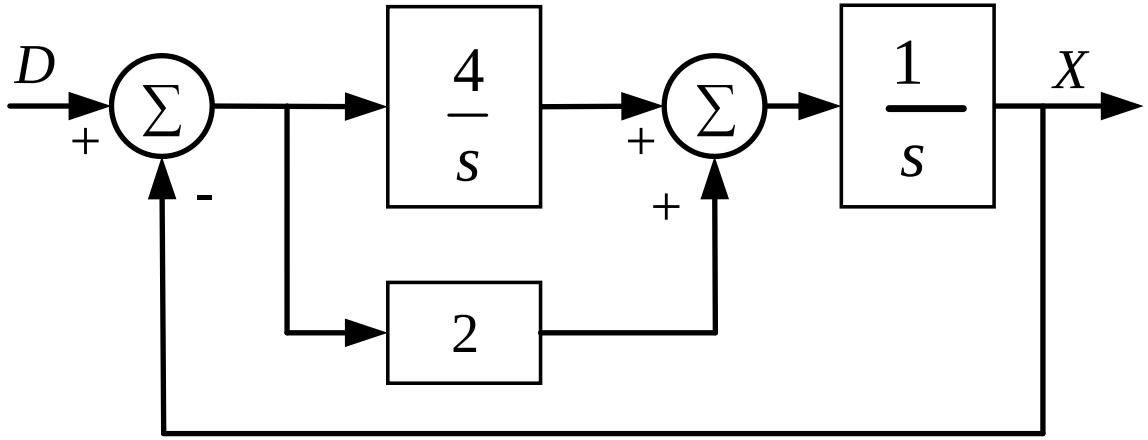
Solution

- Therefore, the time function is given by

$$y(t) = -e^{-2t} + \frac{2}{3}e^{-t} + \frac{1}{3}e^{-4t}$$

Example 6

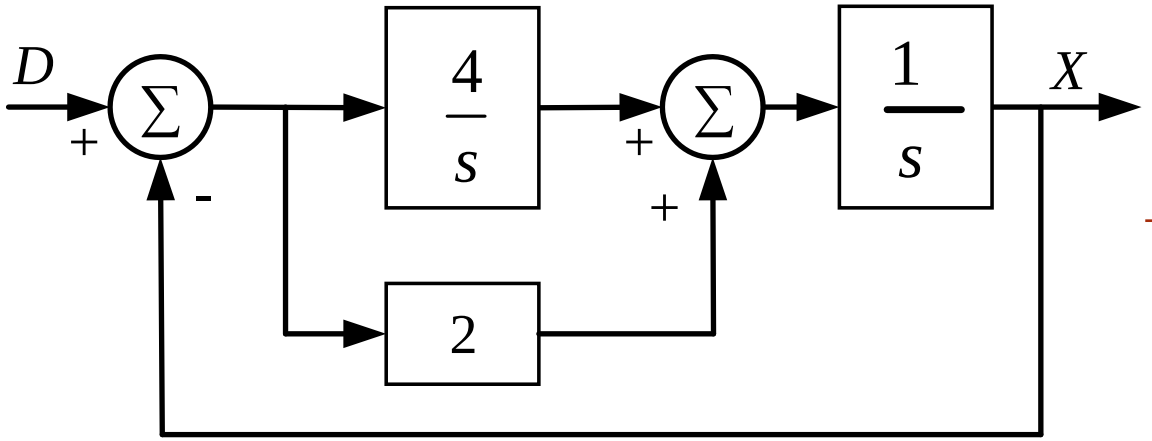
- Find the transfer function of the system shown in the below Fig.

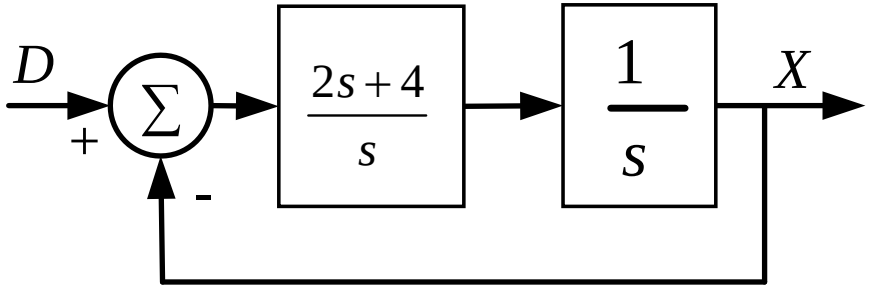


Example 6

Solution

- First we simplify the block diagram by reducing the parallel combination of the controller path.

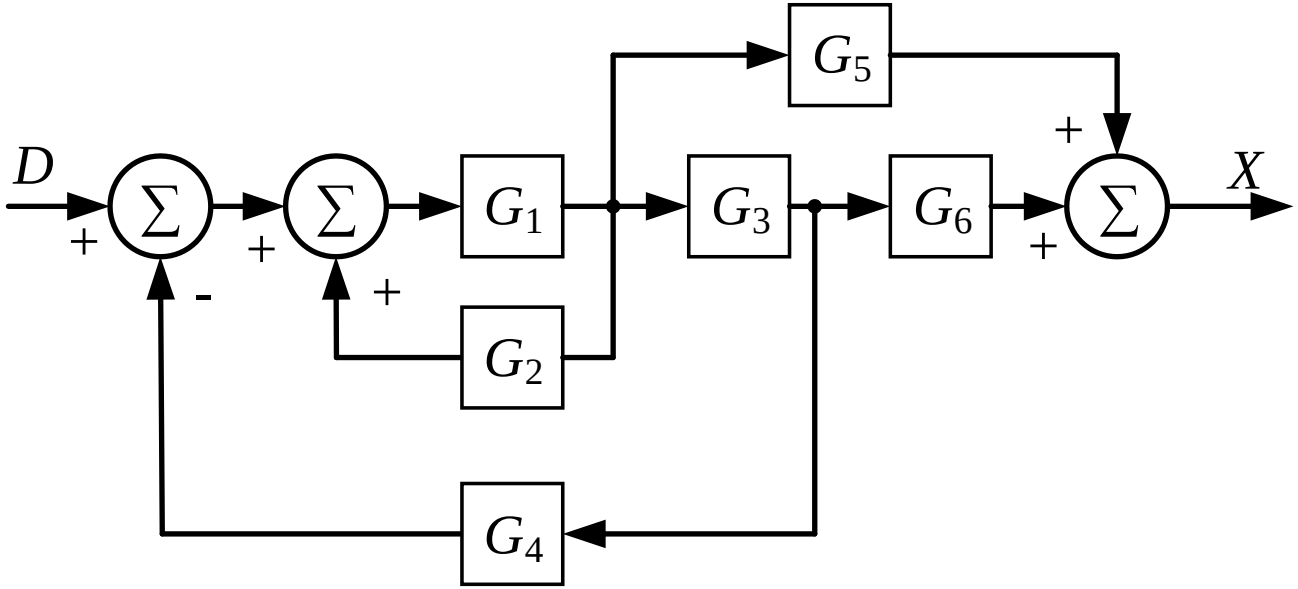




$$T(s)=\frac{X(s)}{D(s)}=\frac{2s+4}{s^2+2s+4}$$

Example 7

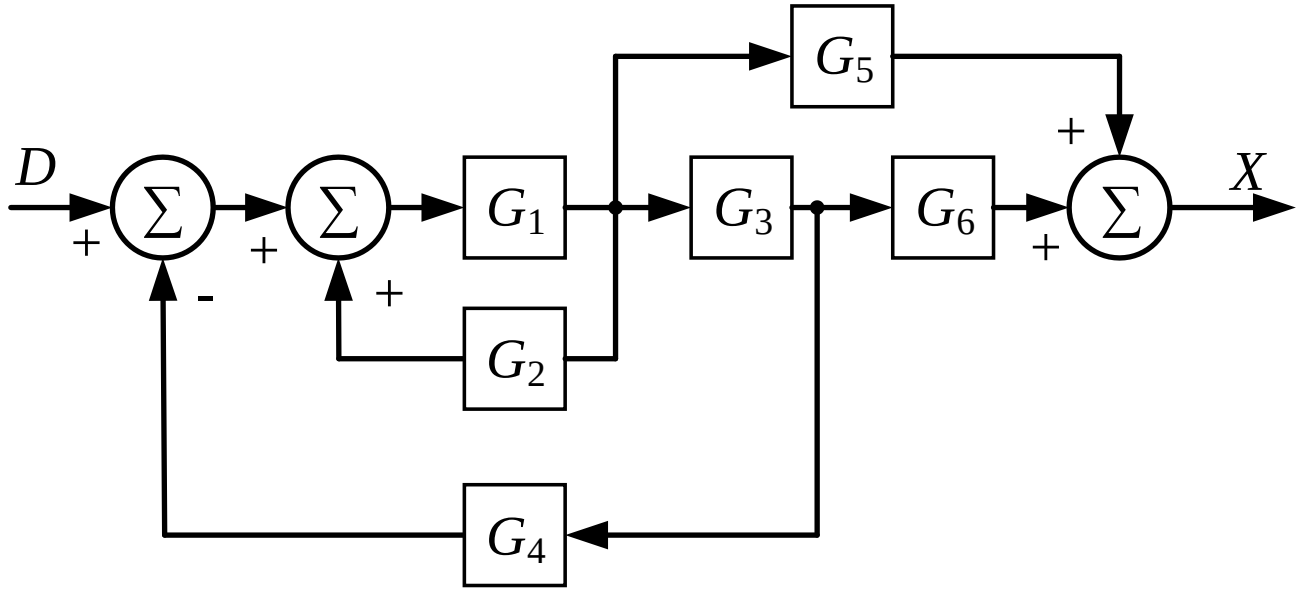
- Find the transfer function of the system shown in below Fig

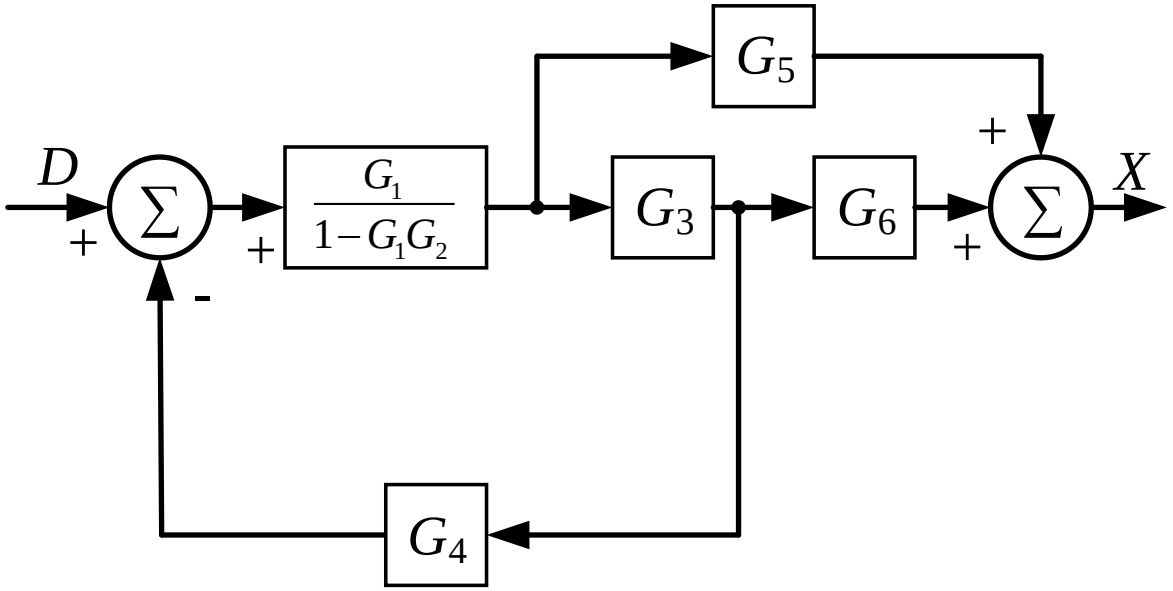


Example 7

Solution

- First we simplify the block diagram. Replace the feedback loop involving G_1 and G_2 by its equivalent transfer function, noting that it is a positive feedback loop.

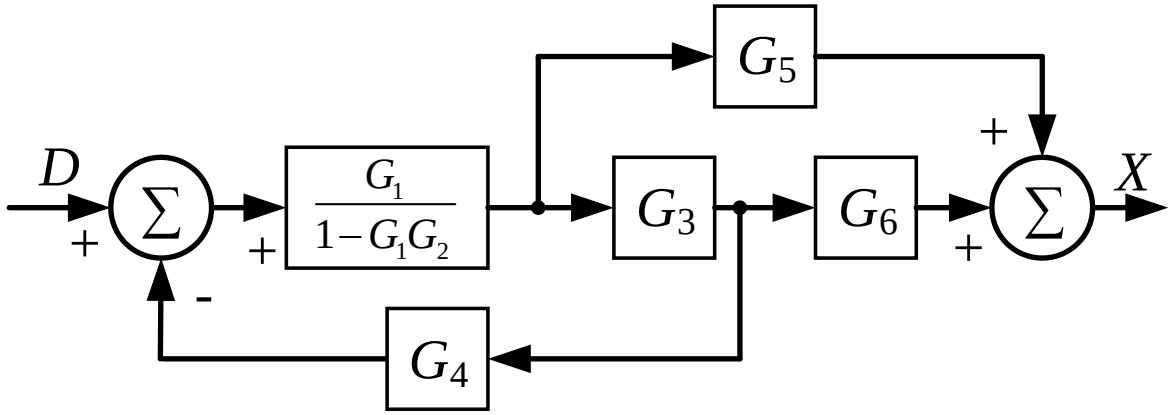


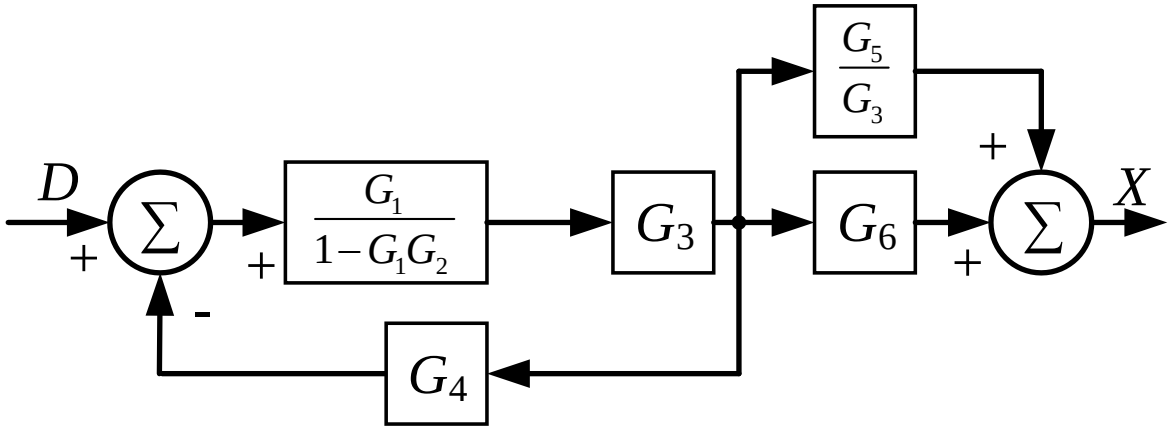


Example 7

Solution

- The next step is to move the pick off point preceding G_3 to its output .

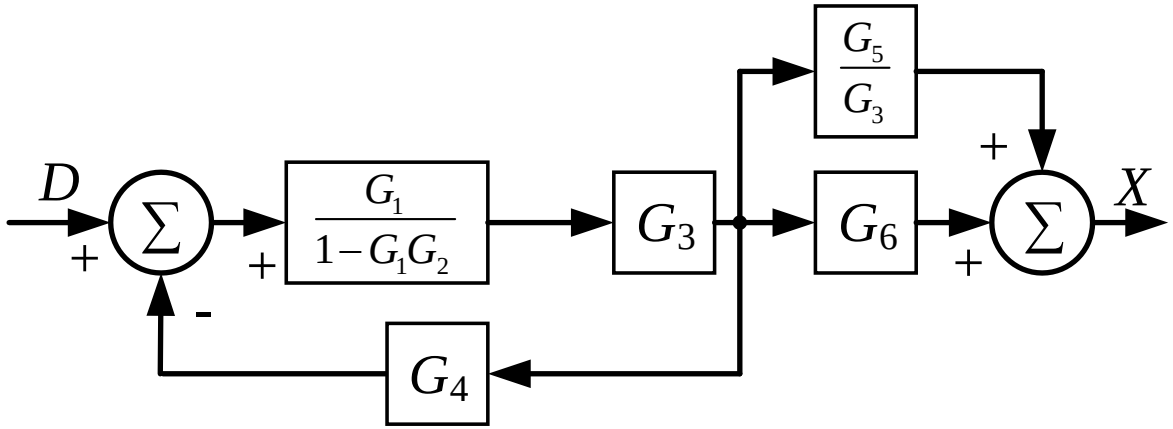




Example 7

Solution

- The negative feedback loop on the left is in series with the subsystem on the right, which is composed of the two parallel blocks G_6 and G_5/G_3. The overall transfer function can be written using all three rules for reduction given by the right Fig.



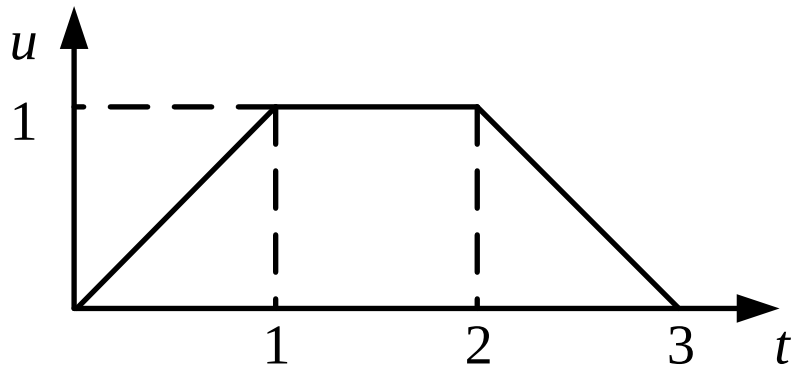
$$T(s)=\frac{X(s)}{D(s)}=\frac{G_1G_3G_6+G_1G_5}{1-G_1G_2+G_1G_3G_4}$$

Example 8

- Consider the standard second-order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

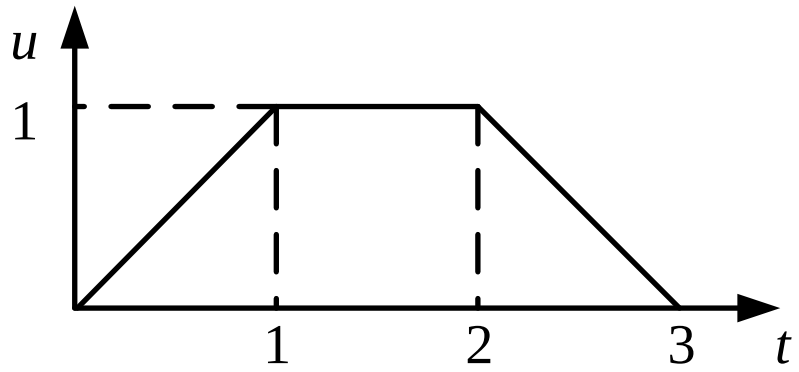
  - **(a)** Write the Laplace transform of the signal in below Fig.
  - **(b)** What is the transform of the output if this signal is applied to $G(s)$?



Example 8

Solution (a)

$$u(t)=\begin{cases} t & 0<t<1 \\ 1 & 1<t<2 \\ 3-t & 2<t<3 \end{cases}$$



$$U(s)=\frac{1}{s^2}-\frac{1}{s^2}\mathrm{e}^{-s}-\frac{1}{s^2}\mathrm{e}^{-2s}+\frac{1}{s^2}\mathrm{e}^{-3s}$$

Example 8

Solution (b)

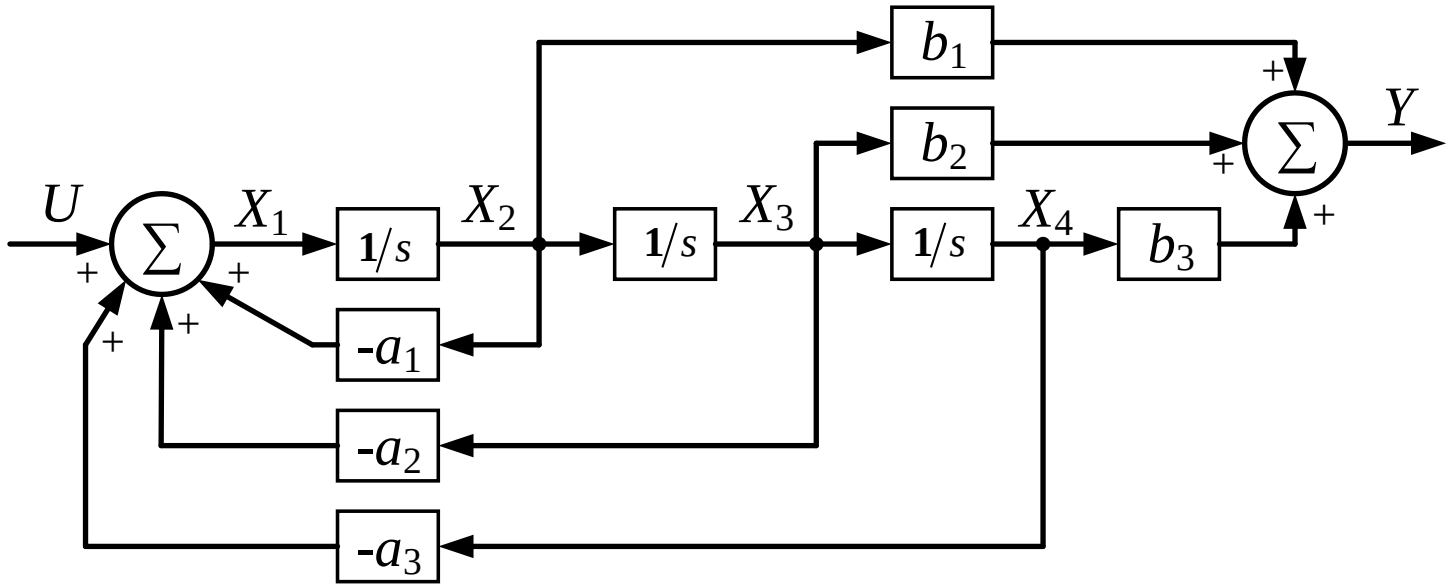
$$G(s)=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$U(s)=\frac{1}{s^2}-\frac{1}{s^2}e^{-s}-\frac{1}{s^2}e^{-2s}+\frac{1}{s^2}e^{-3s}$$

$$Y(s)=G(s)U(s)$$

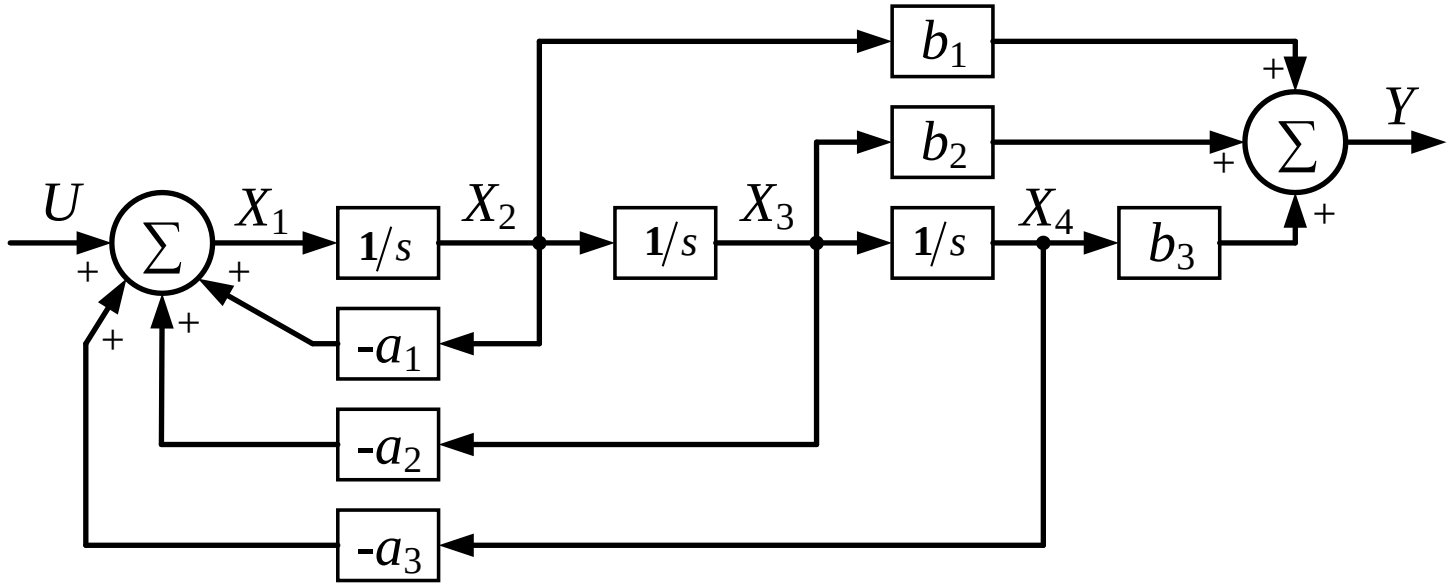
Example 9

- Note that a_i and b_i are constants. Compute the transfer function for this system.



Example 9

Solution



$$X_2 = \frac{1}{s} X_1$$

$$X_3 = \frac{1}{s} X_2 = \frac{1}{s^2} X_1 \qquad X_4 = \frac{1}{s} X_3 = \frac{1}{s^3} X_1$$

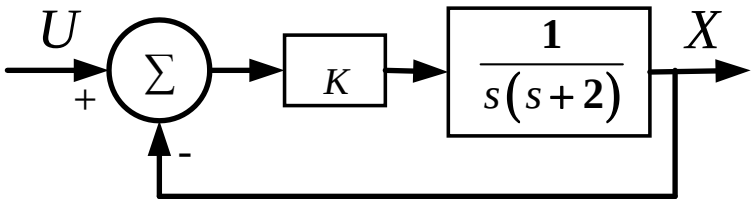
$$X_1 = U - a_1 X_2 - a_2 X_3 - a_3 X_4 = U - \frac{a_1}{s} X_1 - \frac{a_2}{s^2} X_1 - \frac{a_3}{s^3} X_1$$

$$X_1 = \frac{U}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}} = \frac{s^3}{s^3 + a_1 s^2 + a_2 s + a_3} U$$

$$Y = b_1 X_2 + b_2 X_3 + b_3 X_4 = \frac{b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$$

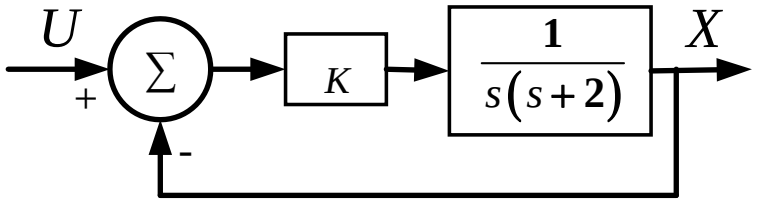
Example 10

- For the unity feedback system, the output $x(t)$ is different in response to a unit step when the gain K of the proportional controller. Find the output $x(t)$ when K=1, 2,10



Example 10

Solution when K=1

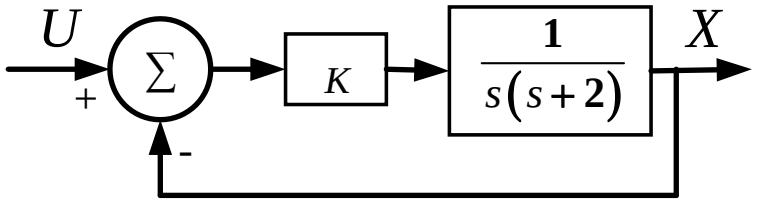


$$X = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

$$x(t) = 1 - (1+t)e^{-t}$$

Example 10

Solution when K=2

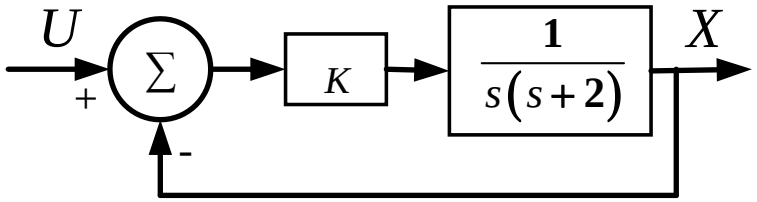


$$X = \frac{1}{s} - \frac{(s+1)}{(s+1)^2+1} - \frac{1}{(s+1)^2+1}$$

$$x(t) = 1 - (\cos t + \sin t)e^{-t}$$

Example 10

Solution when K=10



$$X = \frac{1}{s} - \frac{(s+1)}{(s+1)^2+3^2} - \frac{1}{3}\frac{3}{(s+1)^2+3^2}$$

$$x(t) = 1 - \left(\cos 3t + \frac{1}{3}\sin 3t\right)e^{-t}$$